

Data Warehouses and Data Mining Systems

Report no. 4

Exercise topic:

Phase 4: Physical modelling

Project topic:

Counter Strike skin market analysis

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1 Database Management System

The selected database management system for this project is PostgreSQL. It is a powerful, open-source relational database system known for its reliability, extensibility, and strong support for analytical workloads.

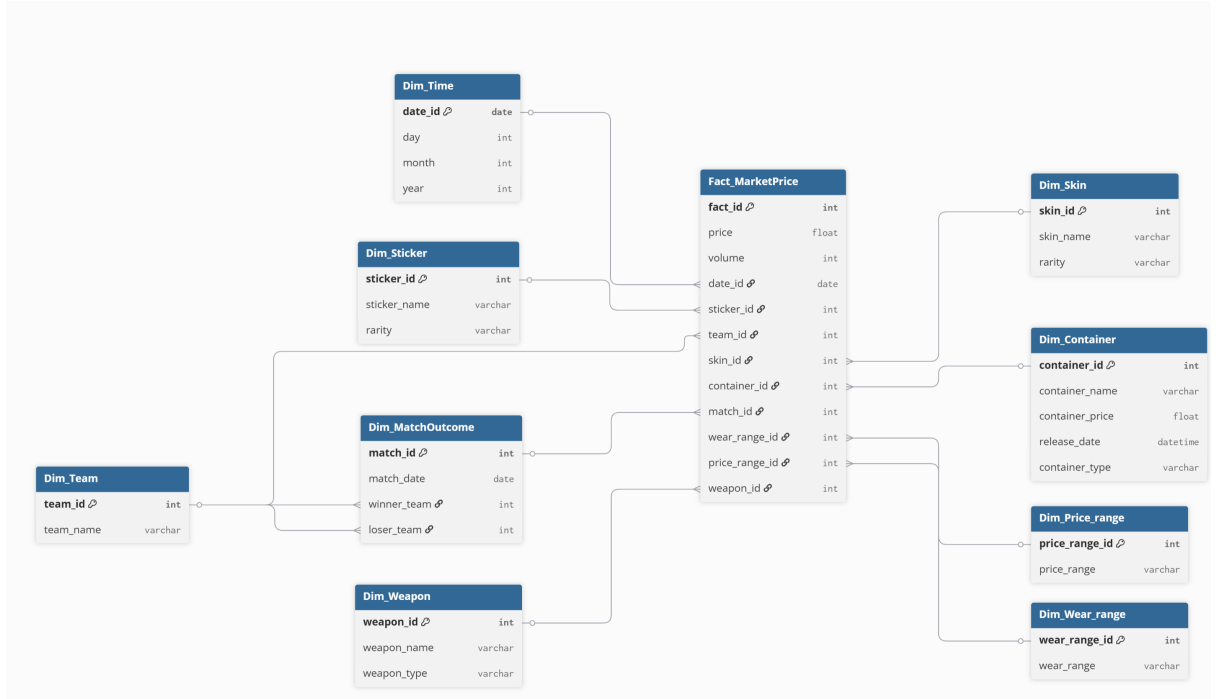
To support the full data warehouse pipeline, additional tools are used:

- Python for ETL processes
- Pandas for data cleaning and transformation
- psycopg2 for communication between Python and PostgreSQL
- Power BI for data visualization and analysis

This combination provides a complete environment for data ingestion, storage, transformation, and reporting.

2 Physical Model

The physical model is derived directly from the logical star schema. It includes one fact table and multiple dimension tables.



2.1 Constraints and Technical Details

- All dimension tables use primary keys for unique identification
- The fact table contains foreign keys referencing all dimensions
- VARCHAR lengths are chosen to balance flexibility and storage efficiency
- Numeric types (FLOAT, INT) are used for measures and identifiers
- Referential integrity is enforced using foreign key constraints

3 DDL SQL Statements

Below are the SQL statements used to create the physical model in PostgreSQL:

```
CREATE TABLE Dim_Time (  
    date_id DATE PRIMARY KEY,  
    day INT,  
    month INT,  
    year INT  
);  
  
CREATE TABLE Dim_Skin (  
    skin_id INT PRIMARY KEY,  
    skin_name VARCHAR(255),  
    rarity VARCHAR(50)  
);  
  
CREATE TABLE Dim_Sticker (  
    sticker_id INT PRIMARY KEY,  
    sticker_name VARCHAR(255),  
    rarity VARCHAR(50)  
);  
  
CREATE TABLE Dim_Weapon (  
    weapon_id INT PRIMARY KEY,  
    weapon_name VARCHAR(255),  
    weapon_type VARCHAR(100)  
);  
  
CREATE TABLE Dim_Container (  
    container_id INT PRIMARY KEY,  
    container_name VARCHAR(255),  
    container_price FLOAT,  
    release_date TIMESTAMP,  
    container_type VARCHAR(100)  
);  
  
CREATE TABLE Dim_Team (  
    team_id INT PRIMARY KEY,  
    team_name VARCHAR(255)  
);  
  
CREATE TABLE Dim_MatchOutcome (  
    match_id INT PRIMARY KEY,  
    match_date DATE,  
    winner_team INT,  
    loser_team INT,  
    FOREIGN KEY (winner_team) REFERENCES Dim_Team(team_id),  
    FOREIGN KEY (loser_team) REFERENCES Dim_Team(team_id)  
);
```

```
CREATE TABLE Dim_Price_range (
    price_range_id INT PRIMARY KEY,
    price_range VARCHAR(50)
);
```

```
CREATE TABLE Dim_Wear_range (
    wear_range_id INT PRIMARY KEY,
    wear_range VARCHAR(50)
);
```

```
CREATE TABLE Fact_MarketPrice (
    fact_id SERIAL PRIMARY KEY,
    price FLOAT,
    volume INT,
    date_id DATE,
    sticker_id INT,
    team_id INT,
    skin_id INT,
    container_id INT,
    match_id INT,
    wear_range_id INT,
    price_range_id INT,
    weapon_id INT,
    FOREIGN KEY (date_id) REFERENCES Dim_Time(date_id),
    FOREIGN KEY (sticker_id) REFERENCES Dim_Sticker(sticker_id),
    FOREIGN KEY (team_id) REFERENCES Dim_Team(team_id),
    FOREIGN KEY (skin_id) REFERENCES Dim_Skin(skin_id),
    FOREIGN KEY (container_id) REFERENCES Dim_Container(container_id),
    FOREIGN KEY (match_id) REFERENCES Dim_MatchOutcome(match_id),
    FOREIGN KEY (wear_range_id) REFERENCES Dim_Wear_range(wear_range_id),
    FOREIGN KEY (price_range_id) REFERENCES Dim_Price_range(price_range_id),
    FOREIGN KEY (weapon_id) REFERENCES Dim_Weapon(weapon_id)
);
```